

Worksheet · Section 3.1–3.2

Variables, Assignment & Lists



The Big Picture

Variables store values; lists store many values under one name. Practice tracing assignments exactly, swapping values, and using AP list indexes (which start at 1!).

Part 1 · Vocab Match

Write the letter of the matching definition on each line.

- | | |
|---|---|
| _____ 1. Variable | A. A table that records each variable's value after every line |
| _____ 2. Assignment ($\leftarrow$) | B. The kind of value: number, string, Boolean, or list |
| _____ 3. Data type | C. Giving a collection of data one name and hiding the details |
| _____ 4. Boolean | D. The position of an element in a list |
| _____ 5. String | E. An ordered sequence of characters |
| _____ 6. List | F. A named place that stores one value at a time |
| _____ 7. Element | G. Stores a value in a variable, replacing the old one |
| _____ 8. Index | H. An extra variable used to hold a value during a swap |
| _____ 9. Data abstraction | I. An ordered sequence of elements stored under one name |
| _____ 10. Empty list | J. One individual value in a list |
| _____ 11. Trace table | K. A list with no elements: [] |
| _____ 12. Temporary variable | L. A value that is either true or false |

Part 2 · Fill It In

Use each word from the word bank once.

index

 $\leftarrow$

most recent

temp

1

error

copy

list

- In AP pseudocode, the assignment operator is _____.
- A variable holds only its _____ value.
- $b \leftarrow a$ gives b a _____ of a 's value.
- The first element of an AP list is at index _____.
- An element's position in a list is its _____.
- Accessing an index larger than `LENGTH(list)` causes an _____.
- Swapping two variables requires a _____ variable.
- Storing 100 test scores under one name uses a _____.

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Practice

Part 3 · Trace Table

```
a ← 6
b ← a - 2
c ← a + b
a ← c * 2
b ← a - c
```

Line	a	b	c
1	6	—	—
2	—	—	—
3	—	—	—
4	—	—	—
5	—	—	—

Part 4 · List Index Practice

```
lst ← ["red", "orange", "yellow",
"green", "blue"]
```

- `lst[1]` → _____
- `lst[LENGTH(lst)]` → _____
- `lst[3]` → _____
- `lst[6]` → _____
- `lst[0]` → _____
- After `lst[2] ← lst[5]`, `lst[2]` → _____

Part 5 · Swap & Rotate

a) What goes wrong with `x ← y` then `y ← x`?

b) Trace: `p ← 1, q ← 2, r ← 3, temp ← p,`
`p ← q, q ← r, r ← temp`

`p` = _____ `q` = _____ `r` = _____

c) Write code to swap `left` and `right`:

Part 6 · Why Lists?

A program stores five scores in `s1 ... s5` and uses many IF statements to find the highest. Give two benefits of using a list instead.



Think:

← is NOT “equals.” Read `x ← x + 1` as “x gets the old x plus 1.” And remember: AP lists start at index 1.

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Exam Practice

Part 7 · Exam Practice

★ Warm-up

1. What is x ?

```
x ← 4
x ← x + x
```

- A. $x + x$
- B. 8
- C. 4
- D. 16

★★ On Level

2. What does `nums` contain?

```
nums ← [3, 6, 9]
nums[2] ← nums[1] + nums[3]
```

- A. [3, 9, 9]
- B. [12, 6, 9]
- C. [3, 12, 9]
- D. [3, 6, 12]

★★ On Level

3. Which is the best variable name for storing whether a game is over?

- A. `isGameOver`
- B. `variable1`
- C. `g`
- D. `x`

★★★ Exam Hard

4. What are `a` and `b`?

```
a ← 5
b ← 3
a ← a + b
b ← a - b
a ← a - b
```

- A. `a = 3`, `b = 8`
- B. `a = 5`, `b = 3`
- C. `a = 8`, `b = 5`
- D. `a = 3`, `b = 5`

★★★ Exam Hard

5. Which TWO statements are true? (*Select two.*)

- A. A list can hold many values under one name.
- B. AP list indexes start at 0.
- C. A variable can hold two numbers at the same time.
- D. On the AP reference sheet, `b ← a` assigns a copy of list `a` to `b`.

Part 8 · Written Response

Explain the difference between `=` in math and `←` in AP pseudocode, using `x ← x + 1` as your example.
